Derivation appendix: 01 — S_3-3HDM scalar parameter space

the λ dictionary, and boundedness from below

1. The $\sqrt{3}$ vacuum alignment $v_2 = \sqrt{3}\, v_1$ is forced by the tadpoles

derived here — §2.1

Claim

$$\tfrac{3}{2}\lambda_4 v_S \left(3v_1^2 - v_2^2\right) = 0$$

Independent checks

- derived from the Lagrangian's three tadpole conditions, not imported
- the corrected [DasDey14] erratum Eq. (9) has the same zero set (section 2.2)

Would be falsified by: a tadpole residual vanishing off $v_2 = \sqrt{3}v_1$ for $\lambda_4 v_S \neq 0$

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2. [DasDey14] Erratum, Phys. Rev. D 91, 039905 (2015) — it corrects Eqs. (9a)–(9c), the tadpole conditions, and nothing else

imported — §2.2

Independent checks

- the corrected (9a)=(9b) forces $v_1 = \sqrt{3}v_2$
- that is our own alignment residual under the documented $v_1 \leftrightarrow v_2$ swap, up to normalisation
- Eq. (4) (boundedness) and Eqs. (36)–(37) (unitarity) are untouched, so section 5 transcribes the final published text

Would be falsified by: a corrected Eq. (9) whose zero set differs from our tadpole residual's

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3. The [GomezBock21] $(a \ldots h) \leftrightarrow$ feynlag λ dictionary, with $e = -\lambda_4$

derived here — §4.2

$$\begin{bmatrix} d & 2\lambda_2 \\ e & -\lambda_4 \\ f & \lambda_6 \\ g & 2\lambda_3 \\ h & 2\lambda_7 \end{bmatrix}$$

Independent checks

- solved from an over-determined linear system, not guessed
- the four solved entries agree with §4.1's term matching
- $e = +\lambda_4$ fails all four published masses
- the sign question vanishes in the $\lambda_4 \to 0$ limit

Would be falsified by: solve() returning no solution, which any error in the potential/tadpole/mass-matrix/rotation chain would cause

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4. The real neutral slice, and the absence of λ_2

derived here — §5.4

Claim

$$V_4 = (\lambda_1 + \lambda_3)\, x^2 + (\lambda_5 + \lambda_6 + 2\lambda_7)\, xy + \lambda_8\, y^2 + 2\lambda_4 \sqrt{y}\, x^{3/2} \cos 3\psi$$

Result

$$\begin{aligned} V =& 2\lambda_4 r_1 r_S \left(r_1^2 - 3r_2^2\right) + \lambda_8 r_S^4 + r_S^2 \left(r_1^2 + r_2^2\right) (\lambda_5 + \lambda_6 + 2\lambda_7) \\ &+ (\lambda_1 + \lambda_3) \left(r_1^2 + r_2^2\right)^2 \end{aligned}$$

Independent checks

- degree-4 homogeneous
- the cubic covariant is $\mathbb{Z}_3$-invariant (falsified on a broken sign)
- $\lambda_4 \to 0$ reduces to a quadratic form in (x, y)
- the $r_2 = 0$ hand slice reproduces it

Would be falsified by: any term outside the $(x, y, \cos 3\psi)$ basis, or λ_2 appearing

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5. [DasDey14] Eqs. (4a)–(4g) boundedness conditions

imported — §5.3

Independent checks

- tested directly against V_4: counterexamples found

Would be falsified by: already falsified in §5.3 as a sufficient condition

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6. The exact boundedness condition on the real neutral slice

derived here — §5.5

Claim

$$\min_{t \geq 0} f(t) \geq 0, \qquad f(t) = \lambda_8\, t^4 + (\lambda_5 + \lambda_6 + 2\lambda_7)\, t^2 - 2|\lambda_4|\, t + (\lambda_1 + \lambda_3)$$

Independent checks

- agrees with 60k-direction brute force on every probe point
- Eq. (4g) is exactly $f(1) > 0$

Would be falsified by: a probe point where the closed form and the direction scan disagree on the sign

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7. The hVV couplings are the CP-even block overlaps with the vacuum direction; h_0 is exactly gauge-phobic

derived here — §8.1

Independent checks

- block overlaps equal the eigenvector projections of the un-rotated 3x3 matrix at 300 points
- the three g^2 sum to 1 to 10^{-12}
- $g^2_{h_0 VV} = 0$ exactly

Would be falsified by: a point where overlap and projection disagree, or the sum is not 1
